Image Reconstruction Under Structured Occlusion: Comparing Robust PCA and Deep Neural Networks

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Abstract

Image inpainting is the task of reconstructing occluded or missing regions of an image, with important applications in face restoration and image recognition under partial visibility. In this project, we compare three fundamentally different approaches for reconstructing occluded facial images: Robust PCA (RPCA), U-Net, and Masked Autoencoder (MAE). Using the Cropped Yale Face Dataset, we simulate occlusions through randomly generated masks and evaluate model performance using pixel-level metrics (MAE, MSE, PSNR) over masked regions, along with SSIM to assess perceptual quality over the full image. Our results show that MAE outperforms both U-Net and RPCA, achieving the lowest reconstruction errors and highest perceptual similarity. While RPCA serves as a reasonable baseline, it is less efficient in producing reconstructions of high perceptual quality. These findings highlight the importance of model architecture in masked image reconstruction and suggest that transformer-based approaches offer a significant advantage in handling structured occlusions.

1. Introduction

Machine learning classification has diverse applications, including medical diagnosis, customer behavior modeling, etc. While deep learning has gained prominence, classical machine learning algorithms remain important due to their interpretability, computational efficiency, and robust performance across various domains.

This project examines the behavior of four classifiers in a binary classification exercise. Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine classifiers were trained on 3 different datasets. By evaluating all four models under identical conditions, this project aims to provide a clear comparison of their strengths, weaknesses, and consistency across domains and sizes of training data.

2. Methods

2.1. Problem Formulation

The dataset we used was the cropped YaleB dataset containing faces of 38 people, with about 64 images of each person in varying lighting conditions. The total data set was 2,414 images. We used random mask occlusion to cover one part of each face with black pixels. The objective was to recover the missing pixels in the masked region first using RPCA, then U-Net, and finally MAE.

2.2. Robust PCA

We used RPCA on the occluded image training set to separate them into a low-rank feature matrix and a sparse matrix. The low-rank matrix represents the main features of faces that are redundant across the images. The sparse matrix contains occlusion, shadows, and other noise. Fitting PCA on the low-rank matrix created a new basis of components to represent the features of clean faces without noise. Next, the test set of occluded images was projected onto that basis. Finally, the test faces were reconstructed using the mean face and weights of the eigenfaces.

2.3. U-Net

We performed reconstruction using a U-Net (Ronneberger, O., et al., 2015) designed with an encoder-decoder structure that enables the model to effectively form maps between images. The occluded images are passed through the encoder which progressively compresses them through a series of 3 x 3 kernels and max-pooling operations. Each convolutional layer generates feature maps from the input image while reducing spatial resolution. The first layer generates 32 feature maps while the third and final layer generates 128 feature maps. The encoder layers capture abstract representations of facial features from the training set.

The decoder layers perform the reconstruction of the images by successively upsampling the feature maps using transposed convolutions, restoring the original spatial dimensions. However, because the input image is progressively compressed through pooling, there is a loss of fine-grained details in the image. Skip connections addresses this issue

by concatenating high-resolution feature maps from earlier encoder layers with the upsampled feature maps in the decoder. This allows the decoder to use the representation of later layers which generally contain higher level semantic information along with the low-level spatial details from earlier layers of the encoder (Adaloglou, N., 2020). This results in more precise and visually coherent reconstructions, particularly for localized structures in images such as facial features.

Our U-net model takes in an occluded image as its input and learns how to predict the clean version of that image. The network outputs a reconstructed image of the same dimensions. Further, to ensure that the model focuses specifically on reconstructing the masked region of the input image, we defined the loss function to be the mean squared error (MSE) over the masked region only. During reconstruction, the final output is formed by combining the model's predictions within the masked region with the original unmasked pixels from the input image. This formulation is an effort to encourage the model to learn meaningful representations for inferring the missing regions of the input face.

2.4. Masked Autoencoder

The Masked Autoencoder (MAE) model used in our project follows a transformer based encoder-decoder architecture adapted for image reconstruction. Each input occluded image is first partitioned into non-overlapping patches of size 8 x 8. Each patch is then flattened into vectors of dimension 64. The patch vectors are linearly projected into a higher-dimensional embedding space of dimension 256 using a learnable patch embedding layer. The sequence of embedded patches is passed into a stack of encoder blocks consisting of multi-head self-attention (with 8 heads) and feedforward layers, repeated over 6 layers as an attempt to capture global contextual relationships within the image. Unlike the U-Net, the MAE encoder stack learns more long-range dependencies between spatially distant regions of the input image.

The decoder is a lightweight multi-layer perceptron that maps each encoded patch representation back to the pixel space. This produces a reconstruction of each patch in the image. The full image is reconstructed by rearranging the predicted patches into their original spatial arrangement. During training, the model receives the occluded image as input and calculates the mean squared error (MSE) loss over the masked region of the image only. Like in the case of the U-Net, this formulation is our attempt to encourage the model to infer the missing regions of the image.

3. Experiments

3.1. Dataset

In our project, we use the Cropped Yale Face Dataset which is widely used for face recognition and reconstruction tasks. This dataset consists of grayscale images of human faces with variations in lighting, facial expressions, illumination, etc. There are multiple images of each person in the dataset, enabling the evaluation of reconstruction performance across changes in appearance. This dataset was a suitable choice for our project because of the relatively consistent alignment of facial features that enables us to focus specifically on reconstructing occluded face images rather than dealing with large variations in pose, background, orientation, etc.

3.2. Preprocessing

All images are loaded from the dataset which is stored as a NumPy array. The images are normalized after converting to float32 so that the pixel values are contained in the range [0, 1]. The normalization is done using the following formula:

$$x_i^{\text{norm}} = \frac{x_i - X_{\min}}{X_{\max} - X_{\min}}$$

Normalizations provide stabilization during training and ensure compatibility with reconstruction losses and evaluation metrics.

3.3. Mask Generation

To simulate missing facial regions, we synthetically generated occlusions using random rectangular masks. For each image the height and width of the mask is randomly chosen to be between 15The occluded image is defined as

$$x_{\text{occ}} = x \odot (1 - M)$$

where M is a binary mask defined as

$$M_{i,j} = \begin{cases} 1 & \text{if } (i,j) \text{ is in the masked region} \\ 0 & \text{otherwise} \end{cases}$$

Here M is a binary mask such that the pixels that are part of the mask have the value 1. Hence, the above formulation creates an image where the masked region is represented as a black rectangle while the rest of the image is intact.

Further, we fix the random seed used to generate the masks to maintain the same occlusion patterns across our experiments, enabling us to make more fair comparisons.

3.4. Train/Test Split

In each of our experiments, we create an 80/20 train/test split with a fixed random seed to ensure comparability. The split

results in train and test datasets with occluded images along with their corresponding masks and ground truth images. This guarantees alignment between the inputs, targets, and mask regions for metric evaluation.

For the U-Net and MAE, we utilize TensorDataset and DataLoader to convert the train and test dataset into a suitable form to pass into the models. For the RPCA, the images were flattened into 1D feature vectors, for the desired input matrix to be of shape: (number of images, number of features).

3.5. Experiment Setup

U-Net and MAE used MSE loss as the training criterion and Adaptive Momentum Gradient Descent (ADAM) as the optimizer. U-Net and MAE were trained with 200 epochs. RPCA was run on the occluded training set. PCA was fit on the resulting low-rank matrix- the clean face subspace-using 300 principal components. Then, the occluded test images were transformed with PCA and inverse transformed to perform the reconstruction of the data. To keep run time for RPCA manageable, the RPCA function used randomized Singular Value Decomposition (SVD) from sklearn to compute the singular values for only the top 100 components instead of the full dimension of the number of training images. The most meaningful variation of the face structure is captured by these 100 components. Capping the components in this way results in some loss in reconstruction accuracy by discarding smaller variations, but it is a necessary change for feasible computing time.

3.6. Evaluation Metrics

In order to evaluate the performance of the three different models on the reconstruction task we use mean squared error, mean absolute error, peak signal-to-noise ratio, and structural similarity index as our evaluation metrics.

3.6.1. Mean Absolute Error (MAE) and Mean Squared Error (MSE)

We calculate the mean absolute error and mean squared error over the masked region of the input images:

$$\text{MAE} = \frac{1}{|M|} \sum_{i \in M} |x_i - \hat{x}_i|$$

$$\text{MSE} = \frac{1}{|M|} \sum_{i \in M} (x_i - \hat{x}_i)^2$$

Here M is the set of masked pixels, x_i is the ground truth value of the pixel and $\hat{x}_i$ is the reconstructed value of the pixel. The range of MAE and MSE values are [0, 1]. We use these metrics to evaluate how accurately the model predicts the masked pixels.

3.6.2. Peak Signal-to-Noise Ratio (PSNR)

The PSNR metric allows us to assess the reconstruction of models in terms of signal fidelity. We compute the PSNR over the masked region using the formula:

$$\text{PSNR} = 10 \log_{10} \left(\frac{MAX^2}{\text{MSE}} \right)$$

Here MAX denotes the maximum possible pixel value which is 1.0 for normalized image vectors. The PSNR is usually expressed in the decibel (dB) scale. While the PSNR does not have a fixed range of values it can take on, a higher PSNR value generally indicates better reconstruction quality. (Poobathy, D., Chezian, R.M., 2014).

3.6.3. Structural Similarity Index (SSIM)

In addition to pixel-wise metrics, we use the Structural Similarity Index (SSIM) to evaluate the perceptual similarity between the ground truth images and the reconstructed images. SSIM measures similarity of images in terms of luminance, contrast, and structural patterns, and was created to address the mismatch between numerical error values and human perception of reconstructed images (Ndajah, P., 2010). This metric was calculated over the entire image to capture structural consistency in the reconstructed image. This choice allows us to evaluate not only the accuracy of the reconstructed pixels, but also how well the reconstructed region integrates with the surrounding context in the image. The range of the SSIM values are [-1, 1] with higher values indicating better perceptual quality.

4. Results

The MAE model achieves the best results across our three models by achieving the lowest error rates with an MSE of 1.68 x 10^{-4} and MAE of 1.12 x 10^{-2}. The MAE also achieves the highest PSNR of 37.7 dB and highest SSIM of 0.982. These results agree with our initial hypothesis that the MAE would be the best performing model. The MAE seems to be highly effective at reconstructing the occluded regions of images while preserving overall image structure.

Similarly, as expected, U-Net also outperforms RPCA on pixel-level metrics, and the SSIM of the U-Net is substantially higher than that of the RPCA. RPCA produces higher numerical reconstruction errors in the masked region and is less effective in preserving the overall structural and perceptual properties of the image. This reflects what is characteristic of PCA reconstruction capability; unique details are diminished and resulting faces appear smoother since reconstruction is a weighted average of eigenvectors.

The quantitative results of our experiments are summarized in the below table.

Model	MSE	MAE	PSNR	SSIM
RPCA	0.08743	0.21634	10.58322	0.67953
U-Net	0.01712	0.07633	17.66613	0.97559
MAE	0.00017	0.01122	37.74922	0.98150

Table 1: Performance of RPCA, U-Net, MAE on reconstruction of occluded facial images.

Examples of our reconstructions can be found in the *Appendix* at the end of this paper.

5. Discussion

5.1. Interpretation

The differences in performance across models can be attributed to their underlying architectures. MAE likely outperforms the others due to its ability to capture global context in an image through self-attention, allowing it to infer missing regions based on long-range dependencies across the image.

In contrast, the U-Net relies on local convolutional operations, generally making it effective at preserving local textures and overall structure, but less capable of accurately reconstructing larger or semantically complex missing regions.

RPCA is limited by its linear formulation, which cannot model the nonlinear variations present in facial data, leading to weaker perceptual quality despite moderate performance on some pixel-wise metrics. However, a visual survey of RPCA reconstructed images suggests that RPCA, with its separation of the sparse noise matrix, accounted for heavily shadowed regions decently well. In cases where a substantial portion of the face was darkened beyond visibility, RPCA recovered a plausible face structure.

5.2. Limitations

Our project has limitations that should be considered when interpreting our results. Firstly, the computational resources available to us constrained both the scale and the number of epochs that could be run in a limited amount of time. The number of epochs may have limited the performance of the data-intensive models such as the MAE and the U-Net. RPCA reconstruction quality may have been decreased with Randomized SVD by limiting the representational capacity of the face subspace. Also, the amount of data used in our project is relatively small compared to modern deep-learning standards, restricting the models' ability to generalize to more diverse and complex datasets.

Additionally, the quality of the dataset is constrained by controlled lighting conditions and limited variability among the images in terms of lighting, pose, orientation, etc. From another perspective, the dataset's quality is also restricted by the representation of people of different genders, races, and ethnicities. Since the dataset does not sufficiently represent diverse facial features, the reconstruction quality may be biased towards certain facial features. These limitations highlight the need for more diverse and representative datasets in order to have more fair and generalizable results in such projects.

6. Conclusion and Future Directions

In our project, we investigated the problem of reconstructing images with masked or occluded regions. We compared Robust PCA, U-Net, and Masked Autoencoder on their ability to recover missing pixel information under a controlled setup. Our experimental results show that there exist clear differences in performance across these methods of performing reconstruction. The transformer-based architecture used in the Masked Autoencoder appears to be the best among the three, achieving the lowest error rates and highest perceptual similarity. U-Net outperforms RPCA especially in terms of perceptual quality of the reconstructed image. RPCA serves as a reasonable baseline while being less efficient at producing reconstructions with high perceptual quality. This is expected due to the limited ability of RPCA to model more complex, nonlinear representations.

These results are significant for real-world image reconstruction and inpainting tasks where there are limitations in the amount of reliable data and computing power. In applications where perceptual quality and smooth integration of reconstructed regions are critical, such as face restoration or occlusion removal, approaches that incorporate higher contextual reasoning, such as MAE, are more suitable. Simpler methods like RPCA may still be useful in constrained settings. However, since RPCA relies on weighted representations of redundant pixels across images, it is limited by the requirement for face images to be similarly aligned in orientation and position. Therefore, nonlinear neural architectures are better-suited for the complexity of real-world visual domains.

We aim to extend this project by scaling up both the dataset and training time, which would enable us to perform more comprehensive evaluations of the models' capabilities. We could further increase the impact of the project by exploring more realistic scenarios where occlusions are irregular or more semantically meaningful. Additionally, we can extend the scope of our project to include more complex model architectures.

In summary, this work highlights the importance of model selection in masked image reconstruction and demonstrates

the advantages of modern transformer-based approaches in achieving both accurate and perceptually coherent reconstructions.

References

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Appendix

6.1. RPCA Reconstructions

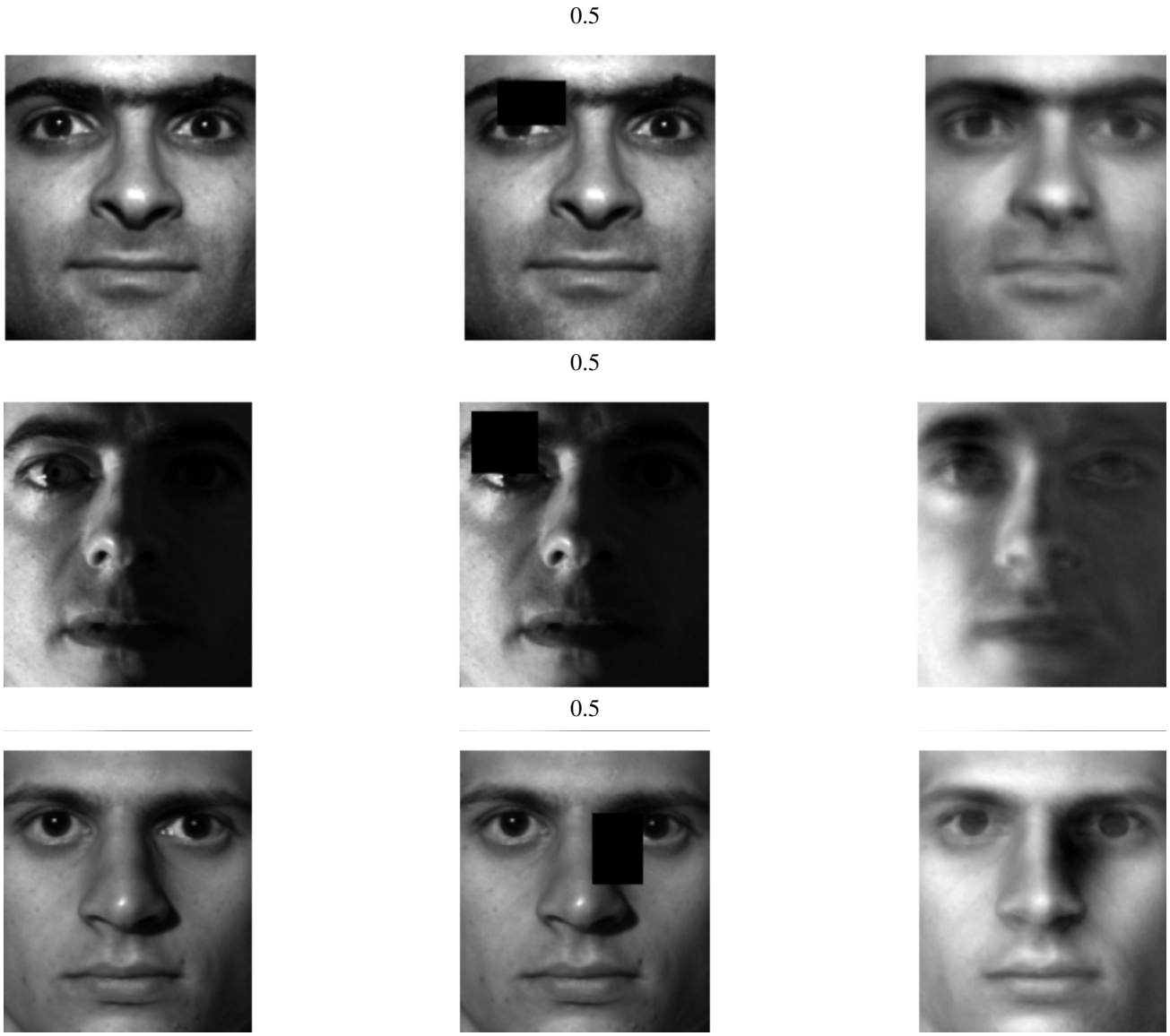

Figure 1: RPCA reconstruction examples.

6.2. U-Net Reconstructions

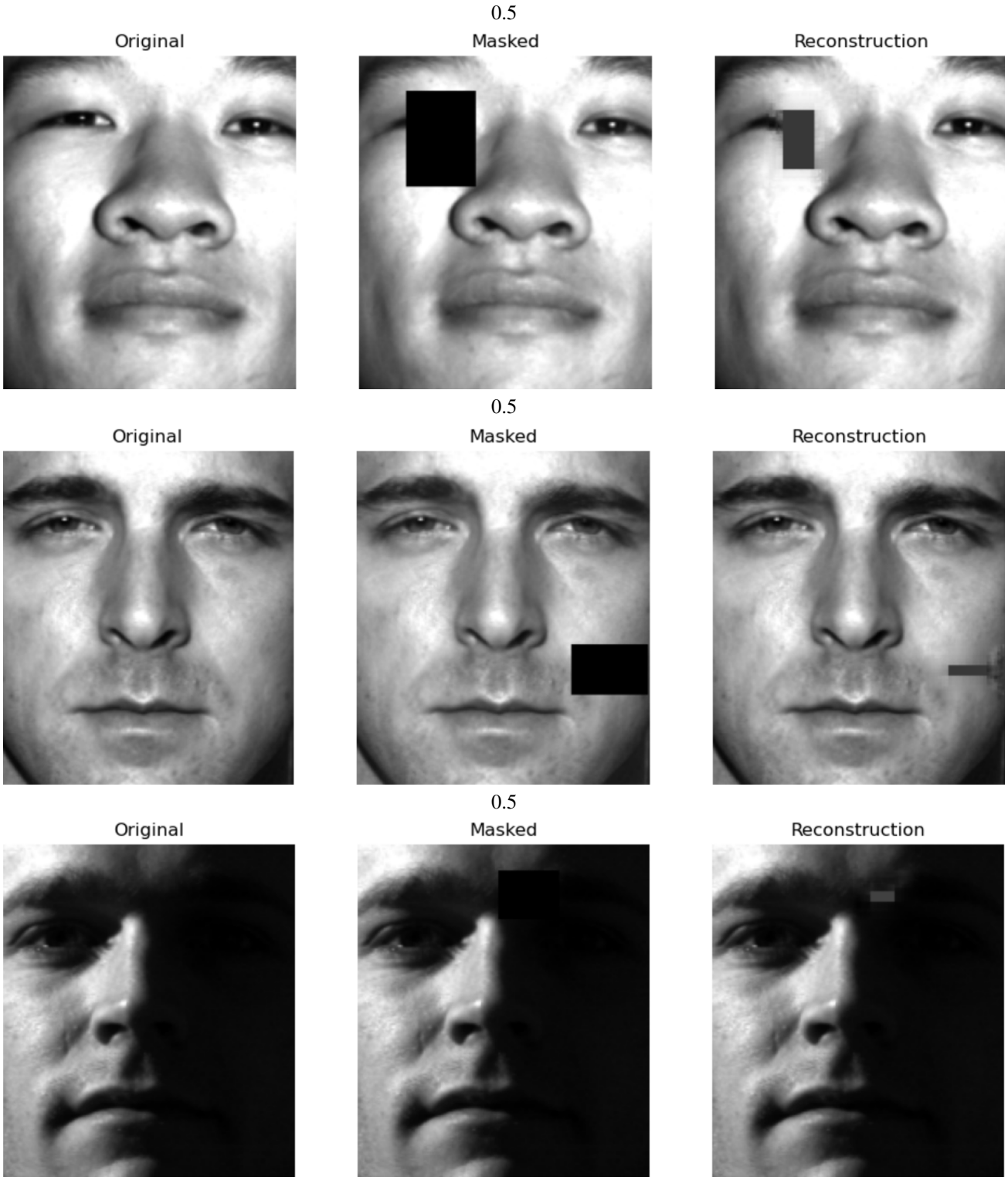

Figure 2: UNet reconstruction examples.

6.3. MAE Reconstructions

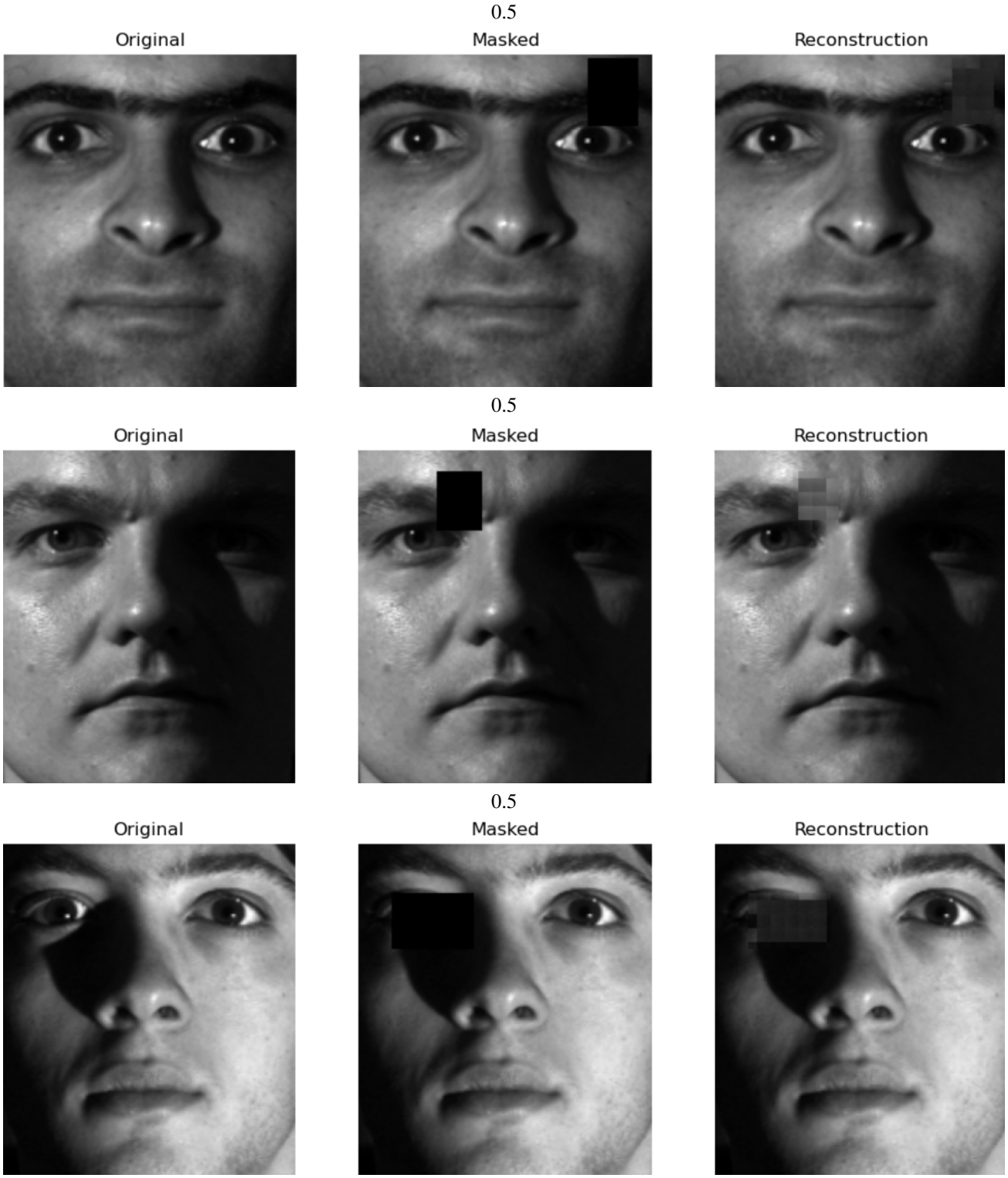

Figure 3: MAE reconstruction examples.